

CONRAD COMPAGNA

Nanaimo, BC, Canada | conradcompagna@gmail.com
language-engine.ai | burmeseneuralreader.com
github.com/conradcompagna

Computational and Digital Humanities | Historian of Empire and Asian Borderlands

I am a PhD historian and digital humanities specialist with a content focus on imperial history and the borderlands linking Burma, China and northeast India. I combine source-language scholarship with full-stack software engineering, applied LLM engineering, NLP model training and statistical, network and spatial analysis. My work includes three completed journal manuscripts, all submitted, a fourth computational-history article in preparation, a source-linked knowledge graph built from the Konbaung chronicles, and two shipped NLP reading platforms of which I am the sole founder and developer. I bring approximately seven years of classroom experience and original course design across global, imperial and Asian history, alongside computational approaches to the humanities.

Education

PhD, History | Birkbeck, University of London | 2020–2026

Dissertation: Layered Empire: Precolonial Continuity, Indigenous Agency, and Hybrid Knowledge on Bengal's Northeast Frontier, 1790–1810.

MSc, History | University of Edinburgh | 2018–2020

BA (Honours), History | University of British Columbia | 2008–2012

Research and manuscripts

- “European Subordination and Burmese Realpolitik: Power Dynamics Across Cultures in the Mid-Eighteenth-Century Irrawaddy Valley” — Journal of Burma Studies, under review.
- “Little Kings, Big Criminals, and Borderlessness on Bengal’s Northern Frontier” — Journal of Borderlands Studies, under review.
- “Empire through the Looking Glass: Late Eighteenth-Century Colonial Knowledge of Burma” — Journal of Imperial and Commonwealth History, under review.
- “The Power of Naming: Tracing Offices, Titles and Patronage in the Konbaung Dynasty through Knowledge Graphs” — in preparation. Combines corpus statistics, network and linguistic analysis, and case studies derived from close reading to examine how the Burmese crown centralized the authority to name and delegate in a setting where it could not establish a monopoly of coercive force.

Computational Approaches to Comparative Empire

Planned monograph extending computational methods to the mountainous highlands between China, Burma and India. Uses large-scale multilingual information extraction and corpus analytics to examine the extent to which European and non-European imperial powers converged on similar methods in their governance and representation of mobile and decentralized highland populations.

Digital humanities projects

Founder and Developer | language-engine.ai | 2025–present

[Source code: github.com/conradcompagna/language-engine](https://github.com/conradcompagna/language-engine)

- Built and shipped a full-stack multilingual NLP reading platform covering more than thirty modern and historical languages, including Japanese, Chinese, Classical Chinese, Korean, Sanskrit, Latin, Greek and Old English.
- Custom-trained transformer models provide tokenization, sentence segmentation, part-of-speech tagging, morphological analysis, lemmatization, dependency parsing and named-entity recognition tailored to each language.
- Created training datasets through synthetic-data generation and human-in-the-loop curation; carried models through training, evaluation and application integration.
- Rewrote the Python model harness to bake dozens of language-specific adapters into static graphs, enabling quantization and CPU deployment of a large research model on a commercial web server.

- Constructed a multi-gigabyte SQLite lexicon from dozens of digitized dictionaries; implemented a prefix- and suffix-aware dynamic-programming word-matching algorithm and language-specific matching rules.
- Integrated the Gemini API for per-token glossing, sentence translation and contextual analysis, including conversation history storage and reinjection, gloss validation and retries, and token usage monitoring.
- Built script-aware transliteration for non-Latin languages and PDF, Word, e-book and HTML rendering, including custom web-capture tooling.
- Owned end-to-end product delivery: user accounts, Stripe billing, analytics, Flask services and deployment using Linux, Gunicorn and Nginx.

Founder and Developer | burmeseneuralreader.com | 2024–present

Source code: github.com/conradcompagna/burmese-neural-reader

- Built and shipped a Burmese reading platform using unigram- and bigram-based statistical language modeling for dictionary segmentation of continuous text without word spaces.
- Trained spaCy models on experimental research datasets for part-of-speech tagging, named-entity recognition and dependency parsing; integrated these analyses into the reading interface through pop-ups and visualization.
- Developed extensive logic for normalizing and reconstructing OCR-damaged historical text, including fuzzy dictionary lookup with BK-trees and Levenshtein edit distance.
- Assembled a large online digitized Burmese lexicon, including Burmese–Pali dictionaries for historical and religious texts.

Developer and Researcher | Konbaung Chronicles Project | 2026–present

Source code: github.com/conradcompagna/konbaung-knowledge-graph

- Built a domain-specific, LLM-powered document extraction pipeline, digitizing, translating and extracting 27,000 triples from a large historical corpus written in eighteenth-century Burmese.
- Designed an ontology of 52 entity categories and 81 relation categories for a knowledge graph stored in Oxigraph and queried through SPARQL and a Flask backend.
- Engineered domain-specific LLM prompts, structured outputs and layered extraction passes across tens of millions of annotation tokens; retained sentence/page provenance so extracted relations remain inspectable against the source.
- Used Gemini vector embeddings for semantic search, entity-resolution candidate discovery and semantic-community clustering.
- Built a Graphology/Sigma browser interface for page, range, focused, thematic, volume and corpus-level network exploration, with relations linked to their supporting evidence and data accessible through an API.
- Applied statistical, network and linguistic analysis to the database for an article in preparation on titles, offices and patronage in the dynasty, connecting the engineered database to original research.

Teaching experience

Beijing No. 101 High School | Beijing | 2017–2022

- Empire in Asia: From the Mongols to Decolonization; Comparative Politics; and European History: Late Middle Ages to the Present.

Beijing No. 8 High School | Beijing | 2023–2024

- Art History: Prehistory to 1500 CE.

New Oriental Foreign Language School | Yangzhou | 2016–2017

- The Americas in Global Perspective.

Editorial experience

Columnist and Editor | China Daily | 2022–2023

- Wrote monthly history-themed columns; edited and ghostwrote news copy; conducted background research, fact-checking and Mandarin/English translation.

Content specialisms

Global and imperial history; South, Southeast and East Asia since 1500; the East India Company; comparative empire; borderlands; indigenous agency; colonial knowledge; digital humanities; natural language processing; computational linguistics.

Technical expertise

NLP: Hugging Face Transformers; PyTorch; TensorFlow; CUDA; transformer and BiLSTM architectures; corpus construction; synthetic datasets; model training and evaluation; runtime quantization; tokenization; morphological analysis; dependency parsing; named-entity recognition; vector embeddings; semantic search.

Software and data: Python; JavaScript; Node.js; HTML/CSS; Flask; SQLAlchemy; SQLite; REST APIs; Linux; Gunicorn; Nginx; dynamic programming; knowledge graphs; RDF; SPARQL; Oxigraph; corpus analysis; data visualization.

Applied LLM engineering: Codex; Claude Code; CLI environments; prompt engineering; context-window management; multi-agent orchestration; Git management; review and QA; LLM API integration; structured outputs.

Languages

Mandarin Chinese: full speaking and reading ability. Burmese and French: reading.

References

References available on request.